EDM101 Day 1: Introduction, Laser Safety, and Type O Project Launch

Students connect music, signal flow, light, safety, and the two-week HOQS Paraflex Type O cabinet project.

EDM101 | CTS Day 25 | Instructor Run Sheet

Electricity becomes experience: sound, light, motion, and safe control. | Monday, August 17, 2026

Core rule

Professional show technology is fun only when the energy is controlled.

Learning targets

- Trace a basic audio or laser signal path from controller to human perception.
- Explain why laser beams require controlled paths, beam termination, and operator responsibility.
- Identify basic speaker-driver parts and explain why an enclosure is part of the sound system.
- Name the two-week Type O cabinet project and record key questions before fabrication begins.

Readiness and materials

Before students arrive	Materials
Set up projector. Place two PA310-8 drivers where students can inspect them. Keep laser unplugged until the safety block and instructor-controlled demo. Mark a clear safe viewing area if the laser is used.	Two Dayton PA310-8 drivers; laptop/projector; slide deck; printed student packets; laser safety test; laser/projector kit if available; targets/beam stop; cones/tape; PPE/eye protection as appropriate; whiteboard markers.

Five-hour run of show

Time	Block	Instructor move
10:00-10:20	Course opening	Introduce EDM101 as music, electricity, sound, light, fabrication, and safe control. Board question: How do we turn electricity into an experience people can hear, feel, and see?

10:20-10:40	Project reveal	Bring out the two PA310-8 drivers. Introduce the HOQS Paraflex Type O 2x12 cabinet as this cohort's main build. Do not over-teach the cabinet yet; create curiosity.
10:40-11:10	Signal chain lesson	Teach source -> processing -> amplifier/output device -> physical energy -> human. Compare audio and laser paths.
11:10-11:45	Laser safety lesson	Cover beam path, termination, reflections, restricted area, operator control, stop-work authority, and no audience scanning. Keep it concrete.
11:45-12:00	Laser safety test	Students complete the safety check independently. Review immediately enough to correct unsafe thinking.
12:00-12:30	Lunch/reset	Reset room. Laser remains controlled by instructor only.
12:30-1:20	Laser lab/demo	Instructor-controlled demonstration. Students trace components and record what changes when control changes. No student operation unless instructor explicitly authorizes.
1:20-1:50	Driver inspection	Students inspect speaker drivers: cone, surround, basket, magnet, terminals, voice-coil location. Ask why the driver alone is not a complete speaker system.
1:50-2:25	Meet the Type O	Show the official HOQS plan or overview image. Students identify drivers, chambers, paths, mouth/opening, panels, and accuracy risks.
2:25-2:45	Project planning	Build a class list: what must happen before music comes out of this cabinet? Capture plan -> measure -> cut -> fit -> assemble -> seal -> wire -> inspect -> test.
2:45-3:00	Exit ticket/reset	Collect exit tickets and safety tests. Reset equipment.

		Preview Day 2: frequency, wavelength, and why cabinets are shaped on purpose.

Instructor notes

- **Keep Day 1 light:** This is orientation and safety, not a full audio-engineering lecture.
- **Laser rule:** If the setup is not clean, the lab becomes a safety planning exercise without energizing the laser. That still counts.
- **Type O framing:** The PA310-8 drivers are temporary build drivers. The cabinet should be built to the official design, then re-drivered later if needed.
- **Tone:** Curiosity first, procedure second, power only after control is proven.

Stop-work triggers

- Beam path crosses an uncontrolled viewing area.
- Reflective object appears near the beam path.
- Students move into the restricted zone without instruction.
- Any student jokes about pointing beams at eyes, phones, aircraft, vehicles, or people.
- Cabinet/drivers are handled roughly or placed where they can fall.

EDM101 Day 1 Laser Safety Test

Complete independently. Use plain language. Safety beats sounding fancy.

EDM101 | CTS Day 25 | Laser Safety Test - Student

Electricity becomes experience: sound, light, motion, and safe control. | Monday, August 17, 2026

1. A laser beam should be aimed through an uncontrolled area as long as nobody is standing there yet. True or false? Explain.

_____

_____

2. Name two things that can create a reflection hazard during laser setup.

_____

_____

3. What is beam termination?

_____

_____

4. Why is audience scanning not allowed in this class lab?

_____

_____

5. What should you do if someone moves into the restricted laser area?

_____

_____

6. Why should phones, shiny tools, mirrors, and watch faces stay out of the beam path?

_____

_____

7. Who controls the laser during class lab?

_____

_____

8. Write one stop-work sentence you could use if something seems unsafe.

_____

_____

EDM101 Day 1 Laser Safety Test Answer Key

Use this to review responses for safe thinking, not perfect wording.

EDM101 | CTS Day 25 | Laser Safety Test - Answer Key
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1. A laser beam should be aimed through an uncontrolled area as long as nobody is standing there yet. True or false? Explain.

Expected answer: False. A beam path must be controlled before the laser is energized. Empty does not mean safe if the area is uncontrolled.

2. Name two things that can create a reflection hazard during laser setup.

Expected answer: Mirrors, phones, watch faces, shiny tools, glass, windows, polished metal, or reflective jewelry.

3. What is beam termination?

Expected answer: A physical target, beam stop, wall surface, or other planned surface that safely receives and ends the beam path.

4. Why is audience scanning not allowed in this class lab?

Expected answer: Because deliberate exposure of people to laser beams requires specialized safety planning, measurement, approvals, and trained operation. This class lab is not an audience-scanning show.

5. What should you do if someone moves into the restricted laser area?

Expected answer: Call stop-work, keep calm, and notify the instructor/operator immediately. Do not attempt to fix it by moving the laser.

6. Why should phones, shiny tools, mirrors, and watch faces stay out of the beam path?

Expected answer: They can redirect concentrated optical energy unpredictably into eyes, people, or uncontrolled areas.

7. Who controls the laser during class lab?

Expected answer: The instructor or specifically authorized operator only.

8. Write one stop-work sentence you could use if something seems unsafe.

Expected answer: Accept any clear stop-work statement: Stop. Beam path is not controlled. Stop. Someone is in the restricted area. Stop. I see a reflection risk.

EDM101 Day 1 Student Packet

Electricity becomes experience: sound, light, motion, and safe control.

EDM101 | CTS Day 25 | Student Print Packet
Electricity becomes experience: sound, light, motion, and safe control. | Monday, August 17, 2026

Today's big question

How do we turn electricity into an experience people can hear, feel, and see?

Signal chain: fill in the job

Stage	What goes in?	What job happens?	What comes out?
Source/controller			
Mixer or processor			
Amplifier or output device			
Speaker or laser projector			
Air, light, and human perception			

Laser safety notes

- A laser is not just a bright light. It concentrates optical energy into a controlled beam.
- Safe use depends on beam path, beam termination, reflections, restricted areas, and operator control.
- Never put eyes, phones, cameras, mirrors, shiny tools, or people into a beam path.
- No audience scanning. No improvising around an energized laser.
- If something feels off, call stop-work and ask.

Speaker-driver inspection

Part	What do you notice?	What job do you think it does?
Cone		
Surround		
Basket/frame		
Magnet		
Terminals		
Voice coil area		

Type O project launch

This cohort will build a HOQS Paraflex Type O 2x12 cabinet. Today we are not trying to understand every chamber. We are trying to ask the right questions before cutting wood.

What I see in the cabinet plan	Question it raises

Exit ticket

1. Trace either an audio or laser signal path from input to human perception.
2. Name one laser-safety rule and explain why it exists.
3. What do you want to understand about the Type O before we build it?

EDM101 Day 1 Lab: Laser Control and Signal Path

Instructor-controlled laser demonstration and systems observation.

EDM101 | CTS Day 25 | Afternoon Lab

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Lab purpose

Students observe how control information becomes visible output, then compare that chain to an audio signal chain.

Safety boundary

- Instructor controls laser power and beam direction.
- Students stay outside the marked restricted area.
- No reflective objects enter the beam path.
- No student changes cabling, aiming, software, or control settings without direct instruction.
- If the safe path cannot be established, complete the lab as a dry setup and planning exercise.

Observation table

System part	What do you see?	What job does it do?	What could fail?
Controller/computer			
Interface/cabling			
Laser projector			
Beam path			
Beam termination/target			

Audio comparison

Laser system	Audio system match	Same idea?
Controller/computer		
Signal/control cable		
Laser output device		
Visible light in room		

Lab close

1. What changed when the control changed?
2. What stayed controlled for safety?
3. How is this similar to sending audio to a speaker?

EDM101 Day 1 Whiteboard Outline

Use this if the slide deck fails or to anchor discussion.

EDM101 | CTS Day 25 | Whiteboard Outline
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Board 1 - Big question

How do we turn electricity into an experience people can hear, feel, and see?

Board 2 - Signal path

SOURCE -> PROCESSING -> OUTPUT DEVICE -> PHYSICAL ENERGY -> HUMAN

Board 3 - Laser safety

- Controlled beam path
- Beam termination
- No reflections
- Restricted area
- Instructor/operator control
- Stop-work authority

Board 4 - Type O project

HOQS Paraflex Type O 2x12 cabinet
Question: Why is the box shaped like that?

Final board sentence

Professional show technology is fun only when the energy is controlled.

EDM101 Day 1 Google Classroom Post

Copy-ready announcement and assignment text.

EDM101 | CTS Day 25 | Google Classroom Post
Electricity becomes experience: sound, light, motion, and safe control. | Monday, August 17, 2026

Announcement

EDM101 starts today. We will connect music, electricity, sound, light, safety, and fabrication. Our main two-week build is a HOQS Paraflex Type O 2x12 cabinet. Today also includes laser safety and a controlled laser demonstration if the setup is safe.

Assignment title

EDM101 Day 1 - Introduction, Laser Safety, and Type O Project Launch

Student directions

1. Complete the laser safety test independently.
2. Use the student packet during the signal-chain and driver-inspection activities.
3. Complete the laser lab observation section during the instructor-controlled demonstration.
4. Complete the exit ticket before class ends.

Turn in

- Laser safety test
- Completed student packet / lab notes
- Exit ticket

Success criteria

Trace the system. Respect the safety boundary. Ask useful questions before building.

EDM101 Two-Week Curriculum Map

Durable course map with Type O cabinet pathway and alternate project options.

EDM101 | CTS Day 25 | Curriculum Map

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Course spine

EDM101 teaches how electricity, signal, power, sound, light, fabrication, and safe operation become a real event-production system. The durable course can run with multiple capstones; this Cohort 6 version uses a HOQS Paraflex Type O 2x12 cabinet as the main build.

Two-week sequence

Day	Focus	Primary artifact or lab
1	Introduction, laser safety, Type O launch	Safety test, system maps, project question list
2	Frequency, wavelength, and human perception	Sound/light frequency comparison; wavelength board math
3	Speaker drivers, impedance, and signal flow	Driver inspection, series/parallel wiring practice
4	Enclosures and why boxes are shaped on purpose	Cabinet plan reading; panel and accuracy checklist
5	Fabrication planning and shop safety	Cut plan, roles, JHA, dry-fit criteria
6	Cabinet assembly and sealing discipline	Build session: layout, fit, fasten, glue/seal documentation
7	Wiring, DSP, crossover, and protection	Terminal cup/wiring plan; conservative test settings
8	Commissioning, measurement, handoff, reflection	Inspection, first sound, handoff sheet, final reflection

Alternate capstone options

- Laser safety and control lab only, if fabrication is unavailable.
- Portable DJ/speaker signal-chain lab using available PA gear.
- Small passive speaker demo box, if shop time or material is limited.
- Lighting/DMX control lab with fixture addressing and scene building.
- Event-system map: power, audio, lighting, safety, and teardown checklist.

Type O project assumptions

- Use official HOQS plan supplied by instructor; do not redraw or modify acoustic geometry for class convenience.
- PA310-8 drivers are acceptable temporary build/test drivers, operated conservatively.
- Permanent re-drivering can happen later without weakening the educational value of the cabinet build.
- Students should learn the difference between works, works safely, and optimized.

Safety doctrine

- No live power without a named job and a visible control boundary.
- No laser operation without a controlled beam path and termination.
- No cabinet test without inspection, wiring verification, and conservative DSP/protection.
- Stop-work is a professional behavior, not a student failure.